

Project-Pizza 🍕 Sales Analysis



NAME: Somnath Phadatare

COURSE: Data Analytics

PROJECT: Sql, Pizza Sales Analysis

This project focuses on analyzing pizza sales data using SQL to extract meaningful insights. It includes queries to calculate total orders, revenue, and identify popular pizza types and sizes. The project also analyzes sales patterns based on time, category, and customer preferences. Advanced SQL concepts like joins, subqueries, and window functions are used. Overall, the project helps in understanding business performance and improving decision-making

```
create database pizza_project;
```

```
use pizza_project;
```

```
show tables;
```

```
-- Baics:
```

```
-- 1. Retrieve the ttal number of orders placed.
```

```
select count(*) as total_orders from orders;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	total_orders			
▶	21350			

```
-- 2. Calculate the total revenue generated from pizza sales.
```

```
CREATE VIEW sales_info AS
```

```
SELECT od.*, (od.quantity * p.price) AS sales
```

```
FROM order_details od
```

```
JOIN pizzas p ON od.pizza_id = p.pizza_id;
```

```
SELECT CONCAT("$", ROUND(SUM(sales),2)) AS total_sale FROM sales_info;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	total_sale			
▶	\$817860.05			

-- 3. Identify the highest-priced pizza.

```
select pt.name, p.price from pizza_types pt
```

```
join pizzas p on pt.pizza_type_id = p.pizza_type_id where p.price = (select Max(price) from pizzas p);
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	name	price			
▶	The Greek Pizza	35.95			

-- 4. Identify the most common pizza size ordered.

```
SELECT p.size,
```

```
SUM(od.quantity) AS total_quantity FROM order_details od
```

```
JOIN pizzas p ON od.pizza_id = p.pizza_id
```

```
GROUP BY p.size ORDER BY total_quantity DESC
```

```
LIMIT 1;
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
	size	total_quantity				
▶	L	18956				

-- 5. List the top 5 most ordered pizza types along with their quantities.

SELECT

pt.name AS pizza_name, SUM(od.quantity) AS total_quantity

FROM order_details od

JOIN pizzas p ON od.pizza_id = p.pizza_id

JOIN pizza_types pt ON p.pizza_type_id = pt.pizza_type_id

GROUP BY pt.name ORDER BY total_quantity DESC

LIMIT 5;

Result Grid	Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
pizza_name		total_quantity		
▶	The Classic Deluxe Pizza	2453		
	The Barbecue Chicken Pizza	2432		
	The Hawaiian Pizza	2418		
	The Pepperoni Pizza	2371		
	The Thai Chicken Pizza	2371		

-- Intermediate:

-- 6. Join the necessary tables to find the total quantity of each pizza category ordered.

SELECT pt.category, SUM(od.quantity) AS total_quantity

FROM order_details od

JOIN pizzas p ON od.pizza_id = p.pizza_id

JOIN pizza_types pt ON p.pizza_type_id = pt.pizza_type_id

GROUP BY pt.category ORDER BY total_quantity DESC;

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
category		total_quantity	
▶	Classic	14888	
	Supreme	11987	
	Veggie	11649	
	Chicken	11050	

-- 7. Determine the distribution of orders by hour of the day.

```
SELECT  EXTRACT(HOUR FROM time) AS order_hour,  
        COUNT(order_id) AS total_orders FROM orders  
  
GROUP BY order_hour ORDER BY order_hour;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
order_hour	total_orders		
9	1		
10	8		
11	1231		
12	2520		
13	2455		
14	1472		
15	1468		
16	1920		
17	2336		
18	2399		
19	2009		
20	1642		
21	1198		
22	663		
23	28		

-- 8. Join relevant tables to find the category-wise distribution of pizzas.

```
SELECT pt.category,  
        SUM(od.quantity) AS total_quantity  
  
FROM order_details od  
  
JOIN pizzas p ON od.pizza_id = p.pizza_id  
  
JOIN pizza_types pt ON p.pizza_type_id = pt.pizza_type_id  
  
GROUP BY pt.category ORDER BY total_quantity DESC;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
category	total_quantity		
Classic	14888		
Supreme	11987		
Veggie	11649		
Chicken	11050		

-- 9. Group the orders by date and calculate the average number of pizzas ordered per day.

SELECT

AVG(daily_total) AS avg_pizzas_per_day

FROM (SELECT o.date,

SUM(od.quantity) AS daily_total

FROM orders o

JOIN order_details od ON o.order_id = od.order_id

GROUP BY o.date) AS daily_orders;

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
avg_pizzas_per_day			
▶ 138.4749			

-- 10. Determine the top 3 most ordered pizza types based on revenue.

SELECT pt.name AS pizza_name,

SUM(od.quantity * p.price) AS total_revenue

FROM order_details od

JOIN pizzas p ON od.pizza_id = p.pizza_id

JOIN pizza_types pt ON p.pizza_type_id = pt.pizza_type_id

GROUP BY pt.name ORDER BY total_revenue DESC

LIMIT 3;

Result Grid	Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
pizza_name	total_revenue			
▶ The Thai Chicken Pizza	43434.25			
The Barbecue Chicken Pizza	42768			
The California Chicken Pizza	41409.5			

-- 11. Calculate the percentage contribution of each pizza type to total revenue.

```
SELECT pt.name AS pizza_name,  
       SUM(od.quantity * p.price) AS total_revenue,  
       ROUND(  
         SUM(od.quantity * p.price) * 100.0 /  
         SUM(SUM(od.quantity * p.price)) OVER (),  
         2) AS percentage_contribution  
FROM order_details od  
  
JOIN pizzas p ON od.pizza_id = p.pizza_id  
  
JOIN pizza_types pt ON p.pizza_type_id = pt.pizza_type_id  
  
GROUP BY pt.name ORDER BY percentage_contribution DESC;
```

Result Grid			
Filter Rows:		Export:	Wrap Cell Content:
	pizza_name	total_revenue	percentage_contribution
▶	The Thai Chicken Pizza	43434.25	5.31
	The Barbecue Chick	Refresh data re-executing the original query	
	The California Chicken Pizza	41409.5	5.06
	The Classic Deluxe Pizza	38180.5	4.67
	The Spicy Italian Pizza	34831.25	4.26
	The Southwest Chicken Pizza	34705.75	4.24
	The Italian Supreme Pizza	33476.75	4.09
	The Hawaiian Pizza	32273.25	3.95
	The Four Cheese Pizza	32265.700000000065	3.95
	The Sicilian Pizza	30940.5	3.78
	The Pepperoni Pizza	30161.75	3.69
	The Greek Pizza	28454.100000000013	3.48
	The Mexicana Pizza	26780.75	3.27
	The Five Cheese Pizza	26066.5	3.19
	The Pepper Salami Pizza	25529	3.12
	The Italian Capocollo Pizza	25094	3.07
	The Vegetables + Vegetable...	24374.75	2.98
	The Prosciutto and Arunula ...	24193.25	2.96

-- 12. Analyze the cumulative revenue generated over time.

```
SELECT order_date,  
  
       daily_revenue,  
  
       SUM(daily_revenue) OVER (ORDER BY order_date) AS cumulative_revenue  
  
FROM ( SELECT o.date AS order_date,  
  
            SUM(od.quantity * p.price) AS daily_revenue  
  
      FROM orders o  
  
     JOIN order_details od ON o.order_id = od.order_id  
  
     JOIN pizzas p ON od.pizza_id = p.pizza_id  
  
    GROUP BY o.date  
  
) AS daily_sales  
  
ORDER BY order_date;
```

Result Grid			
Filter Rows:		Export:	Wrap Cell Content:
	order_date	daily_revenue	cumulative_revenue
▶	2015-01-01	2713.8500000000004	2713.8500000000004
	2015-01-02	2731.8999999999996	5445.75
	2015-01-03	2662.4	8108.15
	2015-01-04	1755.4500000000003	9863.6
	2015-01-05	2065.95	11929.55
	2015-01-06	2428.95	14358.5
	2015-01-07	2202.2000000000003	16560.7
	2015-01-08	2838.3499999999995	19399.05
	2015-01-09	2127.3500000000004	21526.4
	2015-01-10	2463.95	23990.350000000002
	2015-01-11	1872.3000000000002	25862.65
	2015-01-12	1919.0500000000002	27781.7
	2015-01-13	2049.6000000000004	29831.300000000003
	2015-01-14	2527.3999999999996	32358.700000000004
	2015-01-15	1984.8000000000002	34343.500000000001
	2015-01-16	2594.15	36937.650000000001
	2015-01-17	2064.1000000000004	39001.750000000001
	2015-01-18	1976.8500000000001	40978.600000000006
	2015-01-19	2387.1499999999996	43365.750000000001
	2015-01-20	2397.9000000000005	45763.650000000001

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
order_date	daily_revenue	cumulative_revenue	
2015-01-22	2496.7000000000003	50300.900000000001	
2015-01-23	2423.7	52724.600000000006	
2015-01-24	2289.25	55013.850000000006	
2015-01-25	1617.5500000000002	56631.400000000001	
2015-01-26	1884.4000000000003	58515.800000000001	
2015-01-27	2528.0499999999997	61043.850000000001	
2015-01-28	2016	63059.850000000001	
2015-01-29	2045.3000000000002	65105.150000000016	
2015-01-30	2270.3	67375.450000000001	
2015-01-31	2417.85	69793.300000000002	
2015-02-01	3189.1999999999994	72982.500000000001	
2015-02-02	2328.6000000000004	75311.100000000002	
2015-02-03	2614.7999999999997	77925.900000000002	
2015-02-04	2233.8999999999996	80159.800000000002	
2015-02-05	2215.8	82375.600000000002	
2015-02-06	2509.9500000000003	84885.550000000002	
2015-02-07	2237.6500000000005	87123.200000000001	
2015-02-08	2035.0000000000002	89158.200000000001	
2015-02-09	2195.3500000000004	91353.550000000002	
2015-02-10	2056.5	93410.050000000002	

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
order_date	daily_revenue	cumulative_revenue	
2015-02-11	2460	95870.050000000002	
2015-02-12	2158.8000000000006	98028.850000000002	
2015-02-13	2754.5	100783.350000000002	
2015-02-14	2319.1499999999996	103102.500000000001	
2015-02-15	2141.25	105243.750000000001	
2015-02-16	1968.8000000000002	107212.550000000002	
2015-02-17	2121.9	109334.450000000001	
2015-02-18	2642.85	111977.300000000002	
2015-02-19	2030.2500000000002	114007.550000000002	
2015-02-20	2891.15	116898.700000000001	
2015-02-21	2111	119009.700000000001	
2015-02-22	1579.95	120589.650000000001	
2015-02-23	2168.55	122758.200000000001	
2015-02-24	2194.55	124952.750000000001	
2015-02-25	2341.2999999999997	127294.050000000002	
2015-02-26	2261.3	129555.350000000002	
2015-02-27	2857.95	132413.300000000002	
2015-02-28	2539.6000000000004	134952.900000000002	
2015-03-01	1598.55	136551.45	
2015-03-02	2379.05	138930.5	

-- 13. Determine the top 3 most ordered pizza types based on revenue for each pizza category.

SELECT

category, pizza_name, total_revenue

FROM (SELECT pt.category,

pt.name AS pizza_name,

SUM(od.quantity * p.price) AS total_revenue,

RANK() OVER (PARTITION BY pt.category

ORDER BY SUM(od.quantity * p.price) DESC) AS rank_in_category

FROM order_details od

JOIN pizzas p ON od.pizza_id = p.pizza_id

JOIN pizza_types pt ON p.pizza_type_id = pt.pizza_type_id

GROUP BY pt.category, pt.name) ranked_pizzas

WHERE rank_in_category <= 3

ORDER BY category, total_revenue DESC;

Result Grid			
Filter Rows:			
Export: 			
Wrap Cell Content: 			
	category	pizza_name	total_revenue
▶	Chicken	The Thai Chicken Pizza	43434.25
	Chicken	The Barbecue Chicken Pizza	42768
	Chicken	The California Chicken Pizza	41409.5
	Classic	The Classic Deluxe Pizza	38180.5
	Classic	The Hawaiian Pizza	32273.25
	Classic	The Pepperoni Pizza	30161.75
	Supreme	The Spicy Italian Pizza	34831.25
	Supreme	The Italian Supreme Pizza	33476.75
	Supreme	The Sicilian Pizza	30940.5
	Veggie	The Four Cheese Pizza	32265.70000000065
	Veggie	The Mexicana Pizza	26780.75
	Veggie	The Five Cheese Pizza	26066.5